

Use mental addition and subtraction strategies

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $499,999 + 235,678$ mentally. Copy or complete the first step.

2. Calculate $802,000 - 399,998$ mentally.

3. Calculate $675,450 + 99,999$.

4. Miro says: Miro adds 500,000 instead of 499,999 and forgets to adjust. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $499,999 + 235,678$ mentally. Copy or complete the first step.
Answer 735,677.
2. Calculate $802,000 - 399,998$ mentally.
Answer 402,002.
3. Calculate $675,450 + 99,999$.
Answer 775,449.
4. Miro says: Miro adds 500,000 instead of 499,999 and forgets to adjust. Is this correct? Explain.
Answer Compensate by subtracting 1 after adding 500,000.

Use mental addition and subtraction strategies

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $499,999 + 235,678$ mentally.

6. Check this result using an inverse, estimate or known fact: Calculate $675,450 + 99,999$.

2. Calculate $802,000 - 399,998$ mentally.

7. Explain two mental methods for $1,000,000 - 456,789$.

3. Calculate $675,450 + 99,999$.

8. Identify and correct this misconception: Miro adds 500,000 instead of 499,999 and forgets to adjust.

4. Solve this independently and show every stage: Calculate $499,999 + 235,678$ mentally.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $802,000 - 399,998$ mentally.

10. Write and solve a similar question to: Calculate $675,450 + 99,999$.

Teacher answers and checking prompts

1. Calculate $499,999 + 235,678$ mentally.
Answer 735,677.
2. Calculate $802,000 - 399,998$ mentally.
Answer 402,002.
3. Calculate $675,450 + 99,999$.
Answer 775,449.
4. Solve this independently and show every stage: Calculate $499,999 + 235,678$ mentally.
Answer 735,677.
5. Use a second method or representation for: Calculate $802,000 - 399,998$ mentally.
Answer Expected result: 402,002. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $675,450 + 99,999$.
Answer Expected result: 775,449. A valid check must be shown.
7. Explain two mental methods for $1,000,000 - 456,789$.
Answer 543,211. Methods may include complementing or partitioning.
8. Identify and correct this misconception: Miro adds 500,000 instead of 499,999 and forgets to adjust.
Answer Compensate by subtracting 1 after adding 500,000.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $675,450 + 99,999$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use mental addition and subtraction strategies

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Explain two mental methods for $1,000,000 - 456,789$.

6. Create a new example that tests the same idea as: Calculate $675,450 + 99,999$.

2. Prove the answer to this question, rather than only calculating it: Calculate $499,999 + 235,678$ mentally.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $802,000 - 399,998$ mentally.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 775,449.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro adds 500,000 instead of 499,999 and forgets to adjust.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Explain two mental methods for $1,000,000 - 456,789$.
Answer 543,211. Methods may include complementing or partitioning.
2. Prove the answer to this question, rather than only calculating it: Calculate $499,999 + 235,678$ mentally.
Answer Expected result: 735,677. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $802,000 - 399,998$ mentally.
Answer Expected result: 402,002. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 775,449.
Answer One valid reconstruction is: Calculate $675,450 + 99,999$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro adds 500,000 instead of 499,999 and forgets to adjust.
Answer Compensate by subtracting 1 after adding 500,000.
6. Create a new example that tests the same idea as: Calculate $675,450 + 99,999$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.